Ordinary Pietsch measures already characterize injective dominated polynomials

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Source and Question

The source paper is Botelho–Pellegrino–Rueda, *On Pietsch measures for summing operators and dominated polynomials*, arXiv:1210.3332. Section 3 asks for a polynomial analogue of the linear injective Pietsch-measure criterion:

> Given $n \in \mathbb{N}$, is it true that j_p^e is injective if and only if μ is a Pietsch measure for some injective p-dominated n-homogeneous polynomial defined on E?

The source then observes that, for $n \geq 2$, injective n-homogeneous polynomials can occur only in the real odd-degree case, and proves an iff statement with the stronger hypothesis that μ is a *differential* Pietsch measure. The result below answers the stated question in the ordinary Pietsch-measure form in the non-vacuous normed case.

Notation

Let E be a Banach space over $\mathbb{K}$. The dual unit ball $B_{E'}$ is endowed with the weak-star topology, and

$$e : E \longrightarrow C(B_{E'}), \qquad e(x)(\varphi) = \varphi(x),$$

is the canonical isometry. If μ is a regular Borel probability measure on $B_{E'}$, let

$$j_p : C(B_{E'}) \longrightarrow L_p(\mu)$$

be the canonical quotient map and write j_p^e for its restriction to $e(E)$.

For an n-homogeneous polynomial $P : E \to F$, saying that μ is a Pietsch measure for P means that there is a constant C such that

$$\|P(x)\| \leq C \left(\int_{B_{E'}} |\varphi(x)|^p \, d\mu(\varphi) \right)^{n/p} \qquad (x \in E).$$

Proof Intuition

The source uses differential Pietsch measures in order to prove the stronger statement that $j_p^e(e(x)) = j_p^e(e(y))$ implies $P(x) = P(y)$. For the injectivity criterion this is more than is needed. If $j_p^e(e(x)) = j_p^e(e(y))$, then $x - y$ has zero Pietsch seminorm. Ordinary domination applied to $x - y$

gives $P(x-y)=0$. Since a homogeneous polynomial satisfies $P(0)=0$, injectivity of P forces $x-y=0$.

The converse direction is the canonical odd-power construction already implicit in the source: if j_p^e is injective, then

$$x \longmapsto (j_{p/n}^e(e(x)))^n$$

is an injective n-homogeneous polynomial into $L_{p/n}(\mu)$ when the scalars are real and n is odd.

Proposition 1. *Let E be a Banach space over $\mathbb{K}$, let $0<p<\infty$, let μ be a regular Borel probability measure on $B_{E'}$, and let $P \in \mathcal{P}(^nE;F)$ be an injective n-homogeneous polynomial. If μ is a Pietsch measure for P, then $j_p^e : e(E) \to L_p(\mu)$ is injective.*

Proof. Suppose $j_p^e(e(x)) = j_p^e(e(y))$. Since e is linear, this says that $e(x-y)$ is zero in $L_p(\mu)$, hence

$$\int_{B_{E'}} |\varphi(x-y)|^p \, d\mu(\varphi) = 0.$$

Applying the Pietsch domination inequality for P to $x-y$ gives

$$\|P(x-y)\| \le C \left(\int_{B_{E'}} |\varphi(x-y)|^p \, d\mu(\varphi) \right)^{n/p} = 0.$$

Thus $P(x-y)=0=P(0)$. Since P is injective, $x-y=0$, so $x=y$. Therefore j_p^e is injective. $\square$

Theorem 1. *Let E be a real Banach space, let $n \in \mathbb{N}$ be odd, let $p \ge n$, and let μ be a regular Borel probability measure on $B_{E'}$. Then j_p^e is injective if and only if μ is a Pietsch measure for some injective p-dominated n-homogeneous polynomial defined on E.*

Proof. The reverse implication is Proposition 1.

Conversely, assume that j_p^e is injective. Since $p \ge n$, the space $L_{p/n}(\mu)$ is a Banach space. Define

$$P : E \longrightarrow L_{p/n}(\mu), \qquad P(x) = \big(j_{p/n}(e(x))\big)^n.$$

This is a continuous n-homogeneous polynomial. Moreover,

$$\|P(x)\|_{L_{p/n}(\mu)} = \left(\int_{B_{E'}} |\varphi(x)|^p \, d\mu(\varphi) \right)^{n/p},$$

so μ is a Pietsch measure for P with constant 1. By the Pietsch domination theorem for dominated homogeneous polynomials, P is p-dominated.

It remains to check injectivity. If $P(x)=P(y)$, then

$$(j_{p/n}(e(x)))^n = (j_{p/n}(e(y)))^n \quad \text{in } L_{p/n}(\mu).$$

Over the real scalars with n odd, the scalar map $t \mapsto t^n$ is injective, so $j_p(e(x)) = j_p(e(y))$ in $L_p(\mu)$. Since j_p^e is injective, $x=y$. Hence P is injective. $\square$

Limitations and Novelty Check

The theorem is stated for real odd n and $p \geq n$, matching the non-vacuous normed setting isolated in the source. For complex scalars and $n \geq 2$, and for real even n, an injective n-homogeneous polynomial cannot exist because nontrivial roots of unity (or the sign change $x \mapsto -x$) identify distinct points.

Cheap run indexes were searched for arXiv:1210.3332, Pietsch measures, differential Pietsch measures, j_p^e, and injective dominated polynomials. Web searches for exact title and close phrases found the source arXiv record and no later paper with this ordinary-measure refinement. The result should be reviewed as a small strengthening of the source's Proposition 3.4: the differential Pietsch-measure hypothesis is unnecessary for the injective-polynomial criterion.

References

[1] G. Botelho, D. Pellegrino, and P. Rueda, *On Pietsch measures for summing operators and dominated polynomials*, arXiv:1210.3332.